



Air Conditioning Basics

Split · VRF · Packaged · Capacity · Efficiency

• HVAC ENGINEERING • LESSON 04

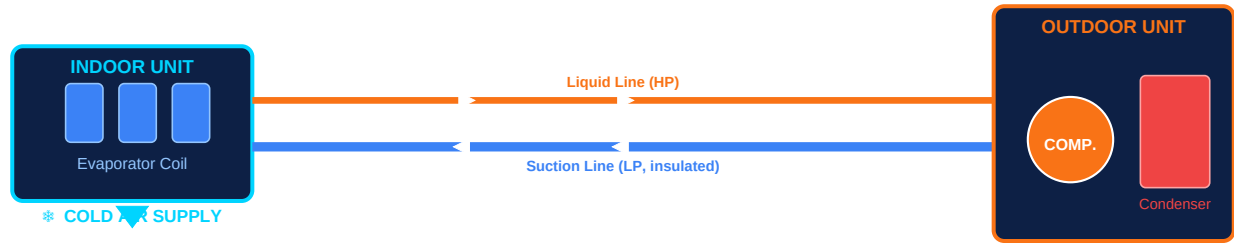
What Air Conditioning Actually Does

Air conditioning does not create cold — it **moves heat**. An AC system absorbs heat energy from inside a building and rejects it outside, leaving the indoor air cooler. This is exactly the refrigeration cycle from HVAC-002, applied to building comfort. Understanding this principle — heat transfer, not cold generation — is the foundation of every AC design decision.

Modern AC systems also control **humidity, air movement, and air quality** — not just temperature. In humid climates, dehumidification is often more important than sensible cooling for occupant comfort.

The Split System — How It Works

Figure 1 — Split AC System: Indoor (Evaporator) ↔ Pipes ↔ Outdoor (Compressor + Condenser) ■ HOT AIR EXHAUST



INDOOR		OUTDOOR	
Component	Location	Function	Key Spec
Compressor	Outdoor unit	Compresses refrigerant vapour; raises pressure and temperature	Type: scroll (residential), inverter-driven for variable speed
Condenser Coil	Outdoor unit	Rejects heat from refrigerant to outside air via fan-forced convection	Fin-and-tube coil; condenser fan 800–1 500 RPM
Expansion Device	Indoor or outdoor unit	Drops refrigerant pressure and temperature; controls flow to evaporator	Electronic TXV on inverter systems; capillary tube on fixed-speed
Evaporator Coil	Indoor unit	Absorbs heat from room air; cools and dehumidifies supply air	Fin-and-tube; drain pan and condensate pump below freezing point
Indoor Fan (FCU)	Indoor unit	Circulates room air over evaporator; controls airflow direction	Cross-flow fan; 3–5 speed settings; auto swing louvers
Refrigerant PIPework	Connecting indoor/outdoor	Carries refrigerant between units; suction line insulated	Copper pipe; max 30 m run; max 15 m height difference (standard)

Inverter vs Fixed-Speed: Fixed-speed compressors run at full power or off — like a light switch. Inverter compressors modulate speed continuously, matching exact load. Inverter systems use 30–50% less energy and maintain tighter temperature control (±0.5°C vs ±2°C).

Air Conditioning System Types

The choice of AC system type depends on building size, number of zones, budget, and the level of individual control required. Each system type represents a different balance of cost, complexity, and capability.



Figure 2 — AC System Types: From Window Unit to VRF

System Type	Capacity	Zones	Best Application	Key Limitation
Window / Through-Wall AC	0.5–2.5 kW	1 room	Single room, budget installs, temporary cooling	Noisy; requires wall/window penetration; no zoning
Single Split System	1–10 kW	1 zone	Residential rooms, small offices, server rooms	One indoor per outdoor; no central control
Multi-Split System	3–14 kW	2–8 zones	Apartments, small commercial — multiple rooms from 1 outdoor unit	All indoor units tied to 1 compressor — partial fault affects all
VRF / VRV System	10–500+ kW	Up to 64 zones	Hotels, offices, retail — simultaneous heating and cooling zones	High upfront cost; complex commissioning; needs specialist service
Packaged Rooftop Unit (RTU)	10–200 kW	1–4 zones	Commercial/industrial buildings with flat roofs and ductwork	Single point of failure; lower efficiency than split at part-load
Chilled Water Fan Coil (CHW FCU)	Unlimited via CHW system	Unlimited	Large commercial/institutional buildings with central plant	High capital cost; requires plant room, cooling tower, pumps

Cooling Capacity — kW, BTU and Tonnage

Unit	Equivalent	When Used
Kilowatt (kW)	1 kW = 3 412 BTU/hr	International standard; used in all modern engineering calculations
BTU/hr	1 BTU/hr = 0.293 W	USA and older UK equipment; still common on product datasheets
Refrigeration Ton (RT)	1 RT = 3.517 kW = 12 000 BTU/hr	Commercial chillers; large systems; used globally in plant sizing
Kcal/hr	1 kcal/hr = 1.163 W	Some older European and Asian equipment specifications

Quick sizing rule: For a well-insulated modern office, allow 80–100 W/m² of cooling load. A 20 m² room needs roughly 1.6–2.0 kW (~7 000 BTU/hr). Always perform a proper load calculation before final equipment selection — rules of thumb can be 30–50% out.

Efficiency Ratings and Energy Performance

Energy efficiency is the most important long-term cost driver in any AC system. Running costs over 15 years often exceed the capital cost of the equipment. Always evaluate life-cycle cost, not just purchase price.

Figure 3 — Cooling EER and Heating COP by System Type



Rating	Full Name	Formula	Notes
EER	Energy Efficiency Ratio	$EER = \text{Cooling output (W)} \div \text{Power input (W) at fixed } 35^{\circ}\text{C outdoor} / 27^{\circ}\text{C indoor}$	Snapshot efficiency; higher = better. EU min: $EER \geq 2.6$. Good: > 3.5
SEER	Seasonal EER (EU standard)	Weighted average EER across European climate year (EU ErP Directive)	More realistic than EER. A++ label $\geq$ SEER 8.5. Used for EU Energy Label
COP (cooling)	Coefficient of Performance	$COP = \text{Cooling kW} \div \text{Electrical kW input (same as EER, dimensionless)}$	$COP = 3.0$ means 3 kW cooling per 1 kW electricity input
COP (heating)	Heat Pump COP	$COP = \text{Heating kW} \div \text{Electrical kW input at } 7^{\circ}\text{C outdoor} / 20^{\circ}\text{C indoor}$	Heat pump COP 3.0–5.5 typical; always > 1.0 (vs electric heater COP = 1.0)
SCOP	Seasonal COP (heating)	Weighted heating COP across heating season (EU ErP Directive)	A+++ heat pump SCOP > 5.1 . Used for EU Energy Label on heat pumps

System Selection Guide

Building Type	Recommended System	Capacity Guide	Key Consideration
Single residential room	Single split inverter (wall-mounted)	1.5–3.5 kW	Choose A++ or higher SEER; verify pipe run ≤ 25 m
Apartment / small house	Multi-split (2–5 indoor units)	5–14 kW total	Size outdoor unit to 90% of sum of indoor units (diversity)
Boutique hotel / guesthouse	VRF with heat recovery	30–150 kW	HR-VRF allows simultaneous heating/cooling in different rooms
Open-plan office (<500 m²)	Ceiling cassette multi-split or VRF	15–50 kW	Cassette gives 360° airflow; good for open plan without ductwork
Retail / supermarket	Packaged rooftop + ductwork or VRF	30–200 kW	Rooftop keeps plant out of lettable area; easy access for service
Hospital / data centre	Chilled water system (centralised)	200 kW–multi-MW	N+1 redundancy critical; precision cooling for data centres

What comes next — HVAC-005: Heating Systems — boilers, heat pumps, underfloor heating, radiators, and how to select and size heating plant for buildings in cold climates.